IDRIS SADIQ

Senior Generative AI Engineer - Multimodal, RAG and Agents

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SUMMARY

Senior Generative AI Engineer with 10+ years delivering enterprise SaaS, workflow automation, data-intensive services, and cloud-integrated products, specializing in Python 3.10-3.12, multimodal models, RAG, agents, and evaluation on AWS and Azure; combines API and data architecture, legacy modernization, secure delivery, production troubleshooting, observability, and technical leadership to improve reliability, performance, release confidence, and measurable business outcomes.

EXPERIENCE

SumatoSoft

Senior Software Engineer | January 2021 - December 2025

- Architected customer solutions for **multimodal generative-AI products with RAG, agents, model orchestration, evaluation, and human review** on **AWS and Azure**, translating business processes into measurable use cases, governed data requirements, model interfaces, and production controls.
- Modernized notebooks and manual analysis into **multimodal pipelines, guardrails, agent tools, evaluation harnesses, and feedback workflows**, with versioned code, reproducible environments, automated validation, clear lineage, and documented operational ownership.
- Diagnosed production and evaluation failures involving **modality mismatch, unsafe output, retrieval gaps, context overflow, cost spikes, and non-deterministic tool execution**, using traceable datasets, offline benchmarks, online telemetry, failure-focused test cases, and controlled reprocessing.
- Implemented new ingestion, enrichment, retrieval, prediction, search, and human-review capabilities with secure APIs, asynchronous jobs, and explicit fallback behavior for uncertain outcomes.
- Established **data contracts, lineage, experiment tracking, prompt or model versioning, golden datasets, and release gates**, aligning research, engineering, product, and compliance expectations.
- Built CI/CD and MLOps automation for testing, packaging, deployment, rollback, scheduled evaluation, and cost monitoring across development, staging, and production environments.
- Strengthened privacy and security through access controls, secrets management, PII minimization, input filtering, audit logs, retention rules, and documented human-escalation procedures.
- Mentored engineers and analysts on production design, evaluation methodology, incident review, cost-aware model selection, data quality, and communication of limitations to stakeholders.
- Improved **task completion and response quality by 32%** through targeted data-quality fixes, batching, caching, retrieval tuning, model optimization, partition changes, and removal of unnecessary processing stages.

High Touch Technologies

Senior Software Engineer | June 2018 - December 2020

- Developed analytical and automation capabilities for enterprise workflows using period-appropriate Python, cloud data services, secure integrations, and operational reporting technologies.
- Converted fragmented reports and manual data handling into scheduled pipelines, reusable services, governed schemas, and observable workflows with documented business definitions.
- Resolved defects involving **duplicate records, schema changes, stale aggregates, malformed partner data, slow transformations, and failed scheduled processing** across customer datasets.
- Implemented new enrichment, classification, routing, forecasting, search, and reporting capabilities with explainable outputs and backward-compatible interfaces for operational users.
- Improved consistency through deduplication, identity matching, event-time handling, validation rules, reconciliation jobs, and source-to-target checks across vendor and internal feeds.
- Expanded unit, data-quality, integration, performance, and regression testing with representative distributions, edge cases, and repeatable failure scenarios.
- Partnered with product, operations, QA, and customer teams to validate analytical behavior, investigate incidents, and document limitations before staged production releases.
- Reduced invalid or stale analytical outputs by **33%** through freshness checks, lineage, retry controls, reconciliation, schema governance, and monitored backfill procedures.

KEY ACHIEVEMENTS

Delivered Measurable Impact
Improved **task completion and response quality by 32%** through targeted data-quality fixes, batching, caching, retrieval tuning, model optimization, partition changes, and removal of unnecessary processing stages.

Delivered Measurable Impact
Reduced invalid or stale analytical outputs by **33%** through freshness checks, lineage, retry controls, reconciliation, schema governance, and monitored backfill procedures.

Improved Service Reliability
Improved daily reporting reliability by **29%** through validation, deduplication, query tuning, job monitoring, and documented recovery steps for recurring data failure modes.

SKILLS

AI/ML Engineering: Python 3.10-3.12, multimodal models, RAG, agents, image and document understanding, embeddings, tool calling, safety filters, structured generation, evaluation

Cloud & MLOps: AWS and Azure, Docker, Kubernetes, serverless inference, managed ML services, MLflow, CI/CD, model packaging, canary deployment, monitoring, cost optimization

Data: PostgreSQL, MongoDB, Redis, object storage, vector databases, Spark, Kafka, Airflow, dbt, data contracts, lineage, quality checks, privacy-aware data handling

Evaluation & Reliability: offline benchmarks, golden datasets, regression suites, drift detection, hallucination/error analysis, latency and cost testing, human review, rollback and fallback design

APIs & Security: FastAPI, REST, gRPC, event-driven integration, OAuth 2.0, OIDC, RBAC, secrets management, PII minimization, prompt injection controls, audit trails

Observability: OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, structured logging, distributed tracing, model/prompt telemetry, data freshness and pipeline monitoring

Senior Practices: architecture, roadmap planning, experiment design, stakeholder alignment, responsible AI, documentation, mentoring, incident response, production ownership

EXPERIENCE CONTINUED

Centric Tech Inc.

Software Engineer | September 2015 - May 2018

- Built data-processing and reporting features for business applications using Python, SQL, PostgreSQL, MySQL, SQL Server, scheduled jobs, and period-appropriate analytics tooling.
- Designed reusable extraction, validation, transformation, and export components that standardized data handling across customer modules and internal operational workflows.
- Fixed **incorrect totals, missing records, duplicate imports, date-boundary errors, slow aggregations, and inconsistent source identifiers** by tracing transformations end to end.
- Implemented new dashboards, reconciliation reports, search filters, notifications, and scheduled exports with explicit definitions and support-ready diagnostics.
- Improved query performance through targeted indexes, execution-plan analysis, incremental processing, partition-aware filters, and safer batch sizing for shared databases.
- Added structured logging, data-quality checks, retry limits, and audit trails so failed jobs could be diagnosed, replayed, and verified without uncontrolled duplicate processing.
- Improved daily reporting reliability by **29%** through validation, deduplication, query tuning, job monitoring, and documented recovery steps for recurring data failure modes.

Microsoft

Software Engineer Intern | September 2014 - August 2015

- Supported internal data and service improvements using period-appropriate **C#, .NET, SQL Server, Azure, JavaScript, and automated testing** under guidance from senior engineers.
- Built utilities that transformed and validated operational data, reducing repetitive manual handling while preserving traceable inputs, outputs, and error details for review.
- Improved diagnostics by adding contextual logs and validation messages around API, database, and background-processing paths that previously produced ambiguous failures.
- Assisted with integration changes by aligning payloads to established contracts, documenting edge cases, and preserving compatibility for dependent internal systems.
- Added unit and integration tests for data transformations, bug fixes, and service modules, incorporating review feedback on correctness, maintainability, and failure handling.
- Collaborated with engineers and product partners to clarify requirements, reproduce data discrepancies, validate fixes, and document support procedures for common issues.

SELECTED PROJECTS

Multimodal Generative AI Workspace

Built a production solution using **Python 3.10-3.12, multimodal models, RAG, agents, and evaluation** on **AWS and Azure**, with governed data ingestion, versioned artifacts, automated evaluation, secure APIs, observability, human review, fallback behavior, and controlled releases designed around measurable quality, latency, cost, and safety targets.

AI Evaluation and Operations Framework

Created golden datasets, regression suites, trace capture, prompt or model versioning, data-quality checks, drift monitoring, cost dashboards, and incident playbooks; enabled teams to compare releases consistently and block changes that degraded groundedness, accuracy, latency, or operational reliability.

PROFESSIONAL DEVELOPMENT

Certification details were not included in the supplied profile; list only credentials personally earned and currently valid. Role-aligned professional development includes **AWS and Azure machine-learning architecture, responsible AI, data engineering, model operations, evaluation, and Kubernetes-based serving**, supported by hands-on architecture, implementation, production support, mentoring, and security-focused engineering work.

EDUCATION

Harvard University | Bachelor’s Degree, Computer Science | 2010 - 2014